

Evidence-governed cognitive-state fusion for task-aware speech screening

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Abstract

Speech-based cognitive screening is scalable, but models trained across heterogeneous tasks and languages can mistake task structure or recording quality for disease evidence. We developed an evidence-governed framework that links source segments to MetricEvidence objects, task-conditioned cognitive StateCards with healthy-reference z-scores, a decomposable supervised prediction (state contributions plus an encoder residual) and an optional large-language-model (LLM) measurement agent. On the PREPARE official test set (n=412; retrospective, because the test set had been viewed previously), the framework with a Whisper encoder branch reached micro-AUC 0.869 and accuracy 0.691 (mean of three seeds), against 0.868 and 0.721 for SpeechCARE: discrimination was at parity, but accuracy remained below SpeechCARE. Adding GPT-5.5 measurement on ASR transcripts did not improve discrimination (micro-AUC 0.868), and a plain GPT reader performed worst (accuracy 0.556, micro-AUC 0.698). The contribution of the framework lies in governance rather than raw accuracy: under prompt rewording the plain agent changed its class in 14.2% of 120 cases versus 1.7% for the framework, evidence groups were highly redundant (effective rank 17.0 of 52 variable metrics), the evidence-group curve reached its minimum out-of-fold log-loss with four groups, and the full framework scored somewhat higher at small training sizes. A historical ten-dataset pilot is reported separately and was not a test of the v2 pipeline. The claims are limited to retrospective research screening.

Keywords: cognitive impairment, speech, evidence governance, clinical decision support, multimodal learning, large language model

1 Introduction

Cognitive impairment is often identified only after functional difficulties have become clinically apparent, whereas specialist assessment, neuroimaging and fluid biomarkers are difficult to deploy as population-scale first-line tests [1, 2]. Speech can be collected repeatedly and remotely, and it reflects performance across language, executive, memory and processing-speed domains rather than a single disease-specific lesion [3]. This distinction defines the intended use of speech artificial intelligence: it may identify people whose cognitive performance warrants further assessment, but speech alone cannot establish Alzheimer pathology, determine aetiology or replace formal clinical evaluation. The clinically relevant target is therefore cognitive-risk screening and referral support with explicit uncertainty and evidence sufficiency.

Speech-based models have progressed from hand-crafted acoustic and linguistic variables to multilingual pretrained representations and adaptive multimodal fusion. SpeechCARE combines acoustic, linguistic and demographic representations with a learned gating network across languages and speech tasks [4]. Such systems can learn whether a case is better represented by audio or text, but a modality weight does not identify which cognitive function was observable, which patient behaviour was abnormal, or whether the apparent signal arose from an interviewer, task duration, automatic speech recognition (ASR) error or recording device. A recent systematic review similarly found persistent gaps in stakeholder evaluation, real-world validation and standardized assessment of explanations in speech-based cognitive models [5]. Predictive discrimination and post-hoc attribution therefore do not by themselves provide a patient-level clinical evidence chain.

A parallel literature has integrated prediction and language-model reasoning in cognitive decision support. Multimodal systems using clinical, cognitive and imaging data have supported differential dementia assessment and clinician decision making [1, 6]. The Multimodal Evidence-Driven Reasoning Framework (MEDRF) combined a hierarchical classifier with retrieval of analogous cases and medical literature, improving three-class external accuracy from 0.706 to 0.753 and producing physician-rated reports [7]. Agentic systems have also operated over longitudinal clinical notes to predict Alzheimer disease or detect documented cognitive concerns [8, 9]. These studies establish the value of combining statistical prediction and language-model reasoning, but they do not solve the measurement problems specific to speech: task-dependent observability, speaker-role separation, recording reliability, confounding, segment-level provenance and permission to translate a low-level feature into a clinician-facing statement.

Here we investigate whether an evidence-governed architecture can make a language-model agent useful, stable and auditable in speech-based cognitive screening. The architecture separates five layers: measurement (deterministic acoustic and linguistic metrics, frozen speech and text encoders, and agent measurement against a fixed knowledge key); evidence (MetricEvidence objects with provenance and permissions); state (task-conditioned StateCards expressed against a healthy reference, with explicit observability); decomposable prediction (per-branch cross-fitted learners combined by out-of-fold stacking); and a governed report that cites only locked, permitted evidence. The agent has three bounded roles: measuring content that deterministic

tools cannot see, offering an independent judgement as one stacking candidate, and auditing evidence quality before a constrained report is written. We evaluate three conditions on the PREPARE official split: acoustic machine learning (A), a plain LLM agent reading the transcript (B), and the framework combined with the agent (C), with GPT-5.5, Claude Opus 5.5 and DeepSeek V4 Pro run under one harness. Evaluation separates predictive performance from governance properties.

Our contributions are as follows.

- *Architecture.* An evidence-governed pipeline that separates measurement, evidence, state, decomposable prediction and governed reporting, so that every clinician-facing statement traces to permitted evidence and every agent influence passes through a learned, bounded channel.
- *Three-condition evaluation.* A comparison of acoustic ML (A), a plain LLM agent (B) and framework plus agent (C) for GPT-5.5, Claude Opus 5.5 and DeepSeek V4 Pro under an identical prompt, schema, reasoning effort, batching and caching, with the same official split and a single use of the test set.
- *Governance properties.* Measurement of (i) stability under prompt rewording, (ii) evidence redundancy and the Pareto curve of evidence groups, (iii) the few-shot learning curve and (iv) a verbatim-continuation memorisation probe, alongside accuracy and discrimination against the published SpeechCARE results.

2 Related work and positioning

2.1 Speech-based cognitive screening

Speech research in cognitive impairment has developed along three technical lines. Early systems relied on hand-crafted acoustic and linguistic variables, including pause timing, speaking rate, lexical diversity, syntactic complexity and semantic content. More recent systems use pretrained speech and language encoders to learn representations that are less dependent on a manually selected feature inventory. A third line combines modalities or tasks through attention, late fusion or learned gates. Across these approaches, speech is best understood as a proxy for performance in several cognitive domains rather than a direct assay of Alzheimer pathology [3]. The distinction is clinically important: a model can support low-burden risk screening while remaining unable to establish disease aetiology.

SpeechCARE is the closest predictive benchmark to the present work [4]. It combines multilingual HuBERT audio representations, multilingual text embeddings and age information, and learns sample-specific modality weights through adaptive gating. Its PREPARE evaluation demonstrates that multilingual and multi-task speech can be integrated without a separate model for every task. However, modality-level weights answer which representation the classifier used, not whether a cognitive state was observable under the current task, which patient behaviour supplied the evidence, or whether the signal was contaminated by interviewer structure, ASR quality, duration or device characteristics. Our work treats those measurement and provenance questions as part of the model interface rather than as post-hoc explanation.

Explainable speech systems have used feature attribution, attention visualization, saliency and natural-language summaries. A recent systematic review found that explanation methods were rarely assessed with patients or clinicians and that real-world validation remained limited [5]. Low-level attribution is also vulnerable to overinterpretation: a spectral coefficient or encoder dimension may aid classification without constituting a clinically reportable mechanism. This motivates our explicit separation among clinical evidence, model-only auxiliary information and quality control.

2.2 Multimodal cognitive decision support

Clinical decision-support studies have combined neuropsychological tests, demographics, structural imaging and electronic health records to classify cognitive stage or dementia aetiology [1, 6]. These systems address a different point in the care pathway from speech screening. They operate after richer clinical information has been collected and can use variables with stronger diagnostic specificity. Their principal methodological challenge is integrating heterogeneous clinical evidence and missing modalities, whereas speech screening must additionally determine whether the elicitation task made a particular behaviour measurable.

MEDRF represents the closest evidence-driven clinical framework [7]. It couples a hierarchical classifier with retrieval of similar cases and medical literature, allowing a language model to refine ambiguous predictions and generate supporting explanations. MEDRF’s hierarchy is diagnostic: cognitive status is followed by disease subtype and aetiology. Our hierarchy is a measurement hierarchy: source segments support metrics, metrics support cognitive states, and states support a screening estimate. The two approaches are therefore complementary. MEDRF demonstrates how retrieved clinical knowledge can support specialist reasoning; the present study asks how evidence should be constructed and governed before a language model is allowed to reason over speech.

2.3 Agentic reasoning and the remaining gap

Language-model agents have been used to synthesize longitudinal notes, retrieve relevant observations and detect documented cognitive concerns [8, 9]. These systems show that an Agent can organize temporally distributed evidence and communicate a decision. They also expose a safety issue: fluent reasoning can cite unsupported facts, conflate a risk factor with an observed impairment or change a model prediction without a reproducible measurement basis. In speech, this risk is amplified because the input contains interviewer speech, task prompts, silence, recording artefacts and ASR errors.

The unresolved gap is therefore not simply multimodal fusion or report generation. It is the absence of a common, machine-verifiable evidence representation linking heterogeneous speech tasks to prediction and clinical communication. We address this gap with four design constraints. First, evidence is conditioned on task, language and speaker role. Second, every metric carries direction, reliability, missingness, confound and report-permission fields. Third, metrics are aggregated into observable cognitive

StateCards with supporting and counterevidence. Fourth, the supervised model and diagnostic Agent operate over the same identifiers, and unsupported Agent proposals are rolled back. This positioning shifts the research question from whether an Agent can produce a plausible cognitive report to whether the complete decision can be reconstructed from permitted patient evidence.

3 Methods

3.1 Study design and intended use

We conducted a retrospective, multi-dataset evaluation of speech-based cognitive-risk screening. The intended use was research screening and referral support. The system was not designed to establish Alzheimer pathology or replace clinical history, neurological examination, validated cognitive testing, imaging or biomarkers. The experimental unit was a participant, subject or public figure according to the source dataset. Repeated recordings from the same identity were aggregated or assigned to the same partition.

3.2 Datasets and held-out partitions

Ten locally available datasets were registered before the final aggregate evaluation: IAEAV, ADReSS 2020, ADReSSo 2021 diagnosis, ADReSSo 2021 progression, PROCESS-2, PREPARE, TAUKADIAL, DementiaBank Pitt, DementiaNet Public-Figures and NCMMSC 2021 AD. Source labels were retained rather than harmonized into a single clinical endpoint. Official subject-disjoint train-test partitions were used when labelled test data were available. Otherwise, deterministic stratified participant-disjoint holdouts of 20–30% were created with a fixed project seed. All three conditions used identical held-out identities within each dataset.

For NCMMSC, only the distributed long-speech track was included; six-second clips and unlabelled recordings were excluded. For Pitt and other CHAT-derived data, patient and interviewer roles were parsed and only patient intervals were used for patient acoustic measurements. IAEAV patient segments were derived from distributed role annotations. For data without speaker roles, the limitation was recorded and role-dependent metrics were marked unavailable. Dataset-specific inclusion criteria and source licenses are provided in Supplementary Table 1. **[AUTHOR INPUT REQUIRED: insert ethics approvals, consent basis and exact data-use agreements.]**

3.3 Audio, transcript and task preprocessing

Audio was resampled to 16 kHz. Processing retained original duration and quality variables, including clipping, a signal-to-noise proxy and role coverage. Recordings were routed by language, task type, source channel and available speaker-role structure. Distributed transcripts were used when available; otherwise ASR used Whisper large-v3-turbo with automatic language selection. ASR outputs retained provenance and were assigned lower reliability when no human-verified transcript was available.

Deterministic metrics, frozen encoders and agent measurements are described in Section 3.4.

Metric availability depended on task, language, transcript quality and speaker roles. Spectral features and quality variables could aid prediction but were not reportable as clinical mechanisms.

Where source age metadata were available, age was encoded into prespecified fixed categories together with an explicit missing-age indicator. The categories were outcome-independent, were used only as optional model context and were never translated into patient-level cognitive evidence or a clinician-facing disease mechanism. Datasets without age metadata did not receive an imputed age category.

3.4 Measurement layer

Measurement produces candidate evidence from four sources. First, deterministic acoustic metrics (pauses, speech-run continuity, rate, intensity, fundamental frequency, spectral descriptors and quality variables) and linguistic metrics (lexical specificity and diversity, utterance length, disfluency, repair, pronoun and content-word ratios) are computed from audio and ASR transcripts. Second, three frozen encoders provide model-only representations: mHuBERT-147 (5 s windows, participant-level mean and standard deviation), gte-multilingual text embeddings, and the Whisper-large-v3-turbo encoder, from which layer 32 is summarised as mean||standard deviation over frames and, in a separate branch, read by a frame-level attention head. Encoder outputs are not translated into cognitive mechanisms.

Third, the LLM agent acts as a measuring instrument. For each recording it receives the pseudonymised transcript, the task type and a fixed knowledge key, for example the Cookie Theft information-unit list, and returns structured measurements of content-unit coverage, coherence and off-topic content, empty words, semantic errors, word-finding difficulty, syntactic complexity and ASR quality. The key, schema and prompt are identical across models, and no labels, demographics or file paths are sent. These measurements are intended to add information that is not already contained in the deterministic metrics or the embeddings; this is the condition under which an agent can contribute at all (Section 3.7). Fourth, test–retest agreement of each agent measurement is recorded and used as a reliability weight.

3.5 MetricEvidence and StateCards

Each measured metric is stored as a MetricEvidence object carrying the value, expected abnormal direction $d_j \in \{-1, 0, 1\}$, task scope, source modality, reliability, missingness, confound tags, report permission and links to source segments. Quality variables have $d_j = 0$ and may lower reliability but cannot support a disease statement. Values are converted to a healthy-reference z-score computed from cognitively healthy training participants only, stratified by task $\times$ language $\times$ age band:

$$z_{ij} = d_j \frac{x_{ij} - \text{median}_{\text{HC}, g(i)}(x_j)}{s_{j, \text{HC}, g(i)}}, \quad (1)$$

where $g(i)$ is the stratum of participant i and s is a robust scale; near-constant references are marked missing, and z is clipped to $[-5, 5]$. A participant thus differs from healthy peers doing the same task in the same language at a similar age, not from a pooled average.

Metrics map to prespecified states (pause and fluency burden, output efficiency, speech continuity, lexical retrieval and specificity, lexical diversity, syntactic complexity, disfluency and repair, content-unit coverage, coherence, among others). A task×state observability mask records which states can be observed for a recording. A state that is not observable, or whose metrics are missing, is treated as missing and never as normal: it is excluded from the weighted mean, lowers state confidence, and may not be reported as unremarkable. For state k , the severity is the reliability-weighted mean of the observable metric z-scores, $S_{ik} = \sum_j w_{jk} r_{ij} z_{ij} / \sum_j w_{jk} r_{ij}$, and the StateCard stores severity, confidence, missing fraction, supporting metrics, counterevidence and local evidence segments.

3.6 Decomposable prediction

Prediction is a stack of per-branch learners. Branches cover acoustic metrics, linguistic metrics, states, stratified z-scores, demographics (age, sex, language, task), mHuBERT, text embeddings, the Whisper layer-32 summary, the Whisper attention head and, where available, agent measurements. Each branch is a regularised multinomial learner whose strength is chosen by five-fold cross-fitting on the training set, yielding out-of-fold (OOF) probabilities for every training participant and fold-averaged probabilities for test participants. A multinomial logistic stacker is fitted on OOF branch log-probabilities only, so that it never sees a branch prediction made by a model trained on the same participant. Encoders, projections and the attention head are fitted inside the folds for the same reason: a head trained on all training data would make the OOF predictions of that branch optimistic and distort the stacker’s weights.

Two decision structures were compared. The flat structure predicts HC, MCI and AD jointly. The two-level structure first separates HC from impaired and then MCI from AD within the impaired group, and recombines the two stages into three-class probabilities. Predictions are averaged over three seeds. Class offsets that trade recall between classes are tuned on the OOF predictions, with a nested estimate (offsets tuned on four fifths of the OOF rows and applied to the remainder) to check that the tuning does not overfit. Because the stacker is linear in branch log-probabilities, the logit decomposes into additive contributions from states, encoder branches and agent features, which supports the contribution table in the report.

3.7 Agent roles and the criterion for agent gain

The agent has three roles. As a *measurer* it produces the content measurements described above. As an *independent judge* it returns class probabilities from the same evidence; this output enters the stack as one more candidate branch and receives a coefficient of zero if it adds nothing. As an *auditor and reporter* it flags ASR errors and unreliable segments, and writes a report from a locked contribution table without changing any number. Writing the supervised mapping from agent outputs to

labels in the stacker, rather than leaving it to the agent, supplies the calibration and population base rates that a plain reader lacks.

Let X denote the information already available to the supervised model, a the agent output and Y the label. An agent can improve prediction only if $I(Y; a | X) > 0$. A plain agent reading the transcript has information in a but no learned map from a to Y ; an agent that re-reads the pauses and voicing already in X has $a \approx f(X)$ and $I(Y; a | X) \approx 0$. We therefore assign the agent the measurements that lie outside X , and test the criterion empirically through the out-of-fold gain of C over the framework without agent measurement.

3.8 Stability bound

With a decomposable logit $\text{logit}(x) = \sum_s \beta_s z_s(x) + \gamma r(x) + b$, an LLM influences the output only through agent-derived state scores z_s ($s \in \mathcal{A}$) and the stacking candidate a with coefficient w_a . If the deterministic parts are unchanged, any drift Δz_s , Δa in these quantities changes the logit by at most

$$|\Delta \text{logit}| \leq \sum_{s \in \mathcal{A}} |\beta_s| |\Delta z_s| + |w_a| |\Delta a|. \quad (2)$$

The bound follows from the structure alone; coefficients are learned from labels, so an unstable or uninformative measurement receives a small weight and the bound tightens. A plain agent (condition B) has no such bound: its drift is the drift of the conclusion.

3.9 Governance invariants

Four invariants hold across the pipeline. Provenance: every metric and statement links to a source segment and a measurement method. Observability: a state may be used only when the task and modality allow it to be observed. Permissions: model-only and quality-only features may influence prediction or reliability but may not appear as disease evidence in the report. Revision closure: any change to evidence recomputes all dependent states and predictions, and stale judgements are not reused. A deterministic validator enforces these properties before a report is compiled.

3.10 Three conditions and providers

Condition A is acoustic machine learning on handcrafted acoustic (and, in A+, text) metrics. Condition B is a plain LLM agent that sees only the pseudonymised transcript and returns HC, MCI and AD probabilities. Condition C is the evidence-governed framework with agent measurement; a variant adds the plain-agent judgement as a stacking candidate. Reference arms add frozen embeddings with demographics (E), the framework without agent (S), and S with the Whisper branch.

GPT-5.5, Claude Opus 5.5 and DeepSeek V4 Pro replace only the agent. They share the same prompt, output schema, reasoning effort, batch size (12 recordings per request) and content-hash caching, so that reruns do not alter results. A refusal,

timeout or schema violation counts as a failure for that recording, and there is no fallback to another model.

3.11 Statistical protocol

We use the PREPARE official split (1,622 training and 412 test participants; 899/216/507 HC/MCI/AD in training and 229/51/132 in test). All model selection, regularisation, stacker choice and class offsets are determined on training OOF predictions. The test set is used once per arm and is reported as the mean of three seeds. Because the official test set had been viewed during earlier development, all test results are retrospective. Reported metrics are accuracy, macro-F1, micro-AUC, log-loss and per-class recall. SpeechCARE values are the published means of ten runs [4]; SpeechCARE fine-tunes its encoders, whereas ours are frozen, and the two protocols differ in repetitions, so we report no formal superiority test. Governance analyses use 120 cases for prompt-rewording stability, test–retest on repeated agent measurement, effective rank and pairwise correlation for redundancy, a greedy out-of-fold log-loss Pareto curve over evidence groups, learning curves on subsampled training sets, and a verbatim-continuation contamination probe. A historical ten-dataset pilot used the earlier pipeline and is reported separately.

3.12 Software and reproducibility

The pipeline is implemented in Python with a versioned, resumable command-line workflow. Model revisions, dataset configuration, split seeds, cache identity, token usage, cost and generated artifacts are recorded for each run. At the 24 September 2026 engineering checkpoint, 1,260 software tests passed and three were skipped; this verifies software behaviour, not clinical performance. The v2 results presented here include GPT-5.5 only; Claude Opus 5.5 and DeepSeek V4 Pro runs are **[pending]**. **[AUTHOR INPUT REQUIRED: insert operating system, Python and package versions, hardware, repository URL, final commit hash and archived release DOI.]**

4 Results

Result-status boundary.

Sections 4.2 to 4.4 report the v2 analysis on PREPARE. The ten-dataset results in Table 1 and Fig. 3c are a locked historical pilot (September 2026) obtained with the earlier pipeline; they motivate routing but are not a test of v2. All PREPARE test results are retrospective.

4.1 Heterogeneous cohorts required task- and source-specific routing

The ten datasets differ in language, task, label set, transcript source and speaker-role availability (Fig. 2). Five datasets provide human or distributed transcripts and five rely on ASR only; speaker roles are available or segmented in five (and per task

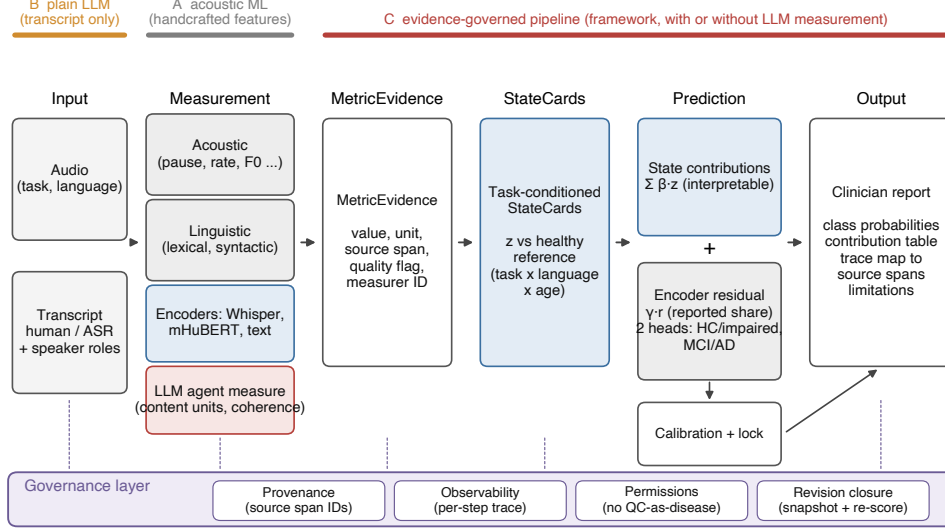


Fig. 1 Evidence-governed speech screening architecture. Audio and transcripts are measured by acoustic, linguistic, encoder and optional LLM-agent measurers; each measurement becomes a MetricEvidence object with source span and quality flag. Task-conditioned StateCards express each metric as a z-score against a healthy reference stratified by task, language and age. A decomposable two-head model combines state contributions with an encoder residual, followed by calibration and locking and a clinician report with a trace map to source spans. A governance layer (provenance, observability, permissions, revision closure) spans the pipeline. Conditions A (acoustic ML), B (plain LLM reader) and C (full pipeline) are indicated.

in PROCESS-2). Label sets range from HC/AD to HC/MCI/AD and a longitudinal decline endpoint. Training sizes are documented here only for PREPARE; for the other datasets the held-out size (6 to 412 participants, 745 in total) is shown and the training size is not reported. Because ASR-only datasets cannot support human-transcript linguistic measurement, and datasets without speaker roles cannot support interaction states, each recording is routed by its channel profile to the states that are observable. Data audits identified three cohorts that could not support an unqualified performance claim: IAEAV (acquisition structure aligned with labels), and DementiaBank Pitt and NCMMSC, where quality-control-only models reached macro-AUROC 0.788 and 0.832, respectively. DementiaNet had six held-out public figures.

4.2 PREPARE: three-condition comparison with SpeechCARE

On the PREPARE official test set (Fig. 3a), accuracy rose from 0.583 (acoustic features) and 0.636 (with text metrics) to 0.673 with frozen embeddings and demographics, 0.687 for the evidence-state framework and 0.691 with the Whisper encoder branch. Micro-AUC followed the same order (0.764, 0.814, 0.857, 0.859 and 0.869). The best framework configuration therefore matched SpeechCARE on discrimination (micro-AUC 0.869 versus 0.868 [4]) but remained below it on accuracy (0.691 versus

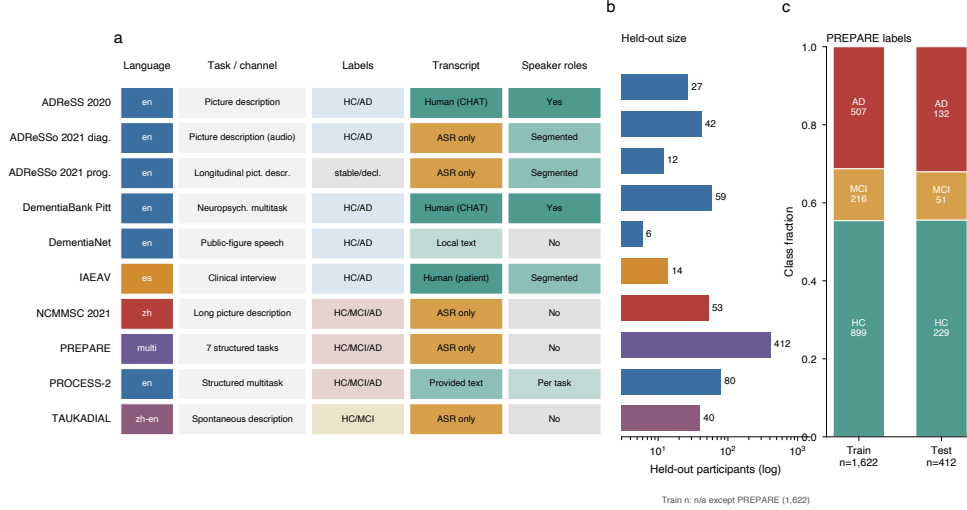


Fig. 2 Data landscape. (a) Language, task or channel, label set, transcript source and speaker-role availability for ten datasets (from the dataset configuration files). (b) Held-out participants per dataset in the historical pilot (log scale); training size is available only for PREPARE (n/a elsewhere). (c) PREPARE class composition in the official training (1,622) and test (412) sets.

0.721). We do not claim superiority over SpeechCARE: the AUC difference is 0.001, SpeechCARE reports a ten-run mean whereas we report three seeds, and the test set was previously viewed. Macro-F1 was 0.592, with recall of 0.88, 0.33 and 0.50 for HC, MCI and AD in the framework arm with Whisper; MCI recall did not exceed 0.44 in any arm (Fig. 3b). Frozen embeddings with demographics already accounted for most of the gain over handcrafted features, so the evidence states added little discrimination beyond the encoders on this dataset; their value is examined in Section 4.4.

4.3 Agent effects

Adding GPT-5.5 measurement to the framework (C) gave accuracy 0.685, macro-F1 0.597 and micro-AUC 0.868, compared with 0.691, 0.592 and 0.869 for the framework with Whisper but without LLM measurement. LLM measurement therefore gave no gain on ASR transcripts; the differences are small and we did not test them formally. A plain GPT reader that saw only the transcript was the worst arm (accuracy 0.556, macro-F1 0.419, micro-AUC 0.698; recall 0.81, 0.25 and 0.23), below even the acoustic baseline in accuracy. In the historical pilot the direct language-model condition (B2) had the lowest macro-AUROC in four of ten datasets (Table 1). The comparison across GPT-5.5, Claude Opus 5.5 and DeepSeek V4 Pro under conditions B and C is [pending] (Fig. 3d); cross-model stability, measurement agreement and the empirical check of the stability bound are likewise [pending].

Table 1 Historical pilot: held-out datasets and three-condition macro-AUROC (locked September 2026 evaluation; earlier pipeline).

Dataset	n	Labels	Language and task	B1	B2	Ours
ADReSS 2020	27	HC/AD	English picture description	0.643	0.530	0.978
ADReSSo diagnosis	42	HC/AD	English audio-only picture description	0.614	0.634	0.857
ADReSSo progression	12	stable/decline	English longitudinal picture description	0.741	0.759	0.556
DementiaBank Pitt [†]	59	HC/AD	English neuropsychological multitask	0.453	0.582	0.897
DementiaNet [‡]	6	HC/AD	English public speech	0.500	0.625	0.500
IAEAV [†]	14	HC/AD	Spanish clinical interview	0.755	0.704	0.939
NCMMSC 2021 [†]	53	HC/MCI/AD	Mandarin long picture description	0.898	0.588	0.910
PREPARE	412	HC/MCI/AD	Multilingual structured tasks	0.728	0.584	0.811
PROCESS-2	80	HC/MCI/AD	English structured multitask	0.737	0.756	0.862
TAUKADIAL	40	HC/MCI	Mandarin–English spontaneous speech	0.459	0.625	0.516

B1, conventional acoustic machine learning; B2, direct transcript-based language model; HC, cognitively healthy control; MCI, mild cognitive impairment; AD, Alzheimer disease label as distributed by the source dataset. [†]Result downgraded to a bias audit because of acquisition or quality-control confounding. [‡]Exploratory engineering check because of the very small held-out sample.

4.4 Governance properties

Stability.

Under prompt rewording in 120 cases, the plain agent changed its predicted class in 14.2% of cases and the framework in 1.7%; mean absolute probability change was 0.062 versus 0.0085 (Fig. 4a). LLM measurements were reproducible on repeat, with test–retest Spearman correlation of 0.82 for information units and for empty-word ratio, 0.79 for word finding, 0.73 for impairment support and 0.62 for coherence (Fig. 4b).

Redundancy and Pareto.

Of 55 metrics, three were constant, 14 pairs had $|\rho| \geq 0.9$ and the effective rank of the 52 variable metrics was 17.0 (Fig. 4c). Adding evidence groups greedily, out-of-fold log-loss fell from 0.723 with one group to a minimum of 0.7065 with four groups (Whisper, audio embedding, text embedding and stratified z) and then drifted upward

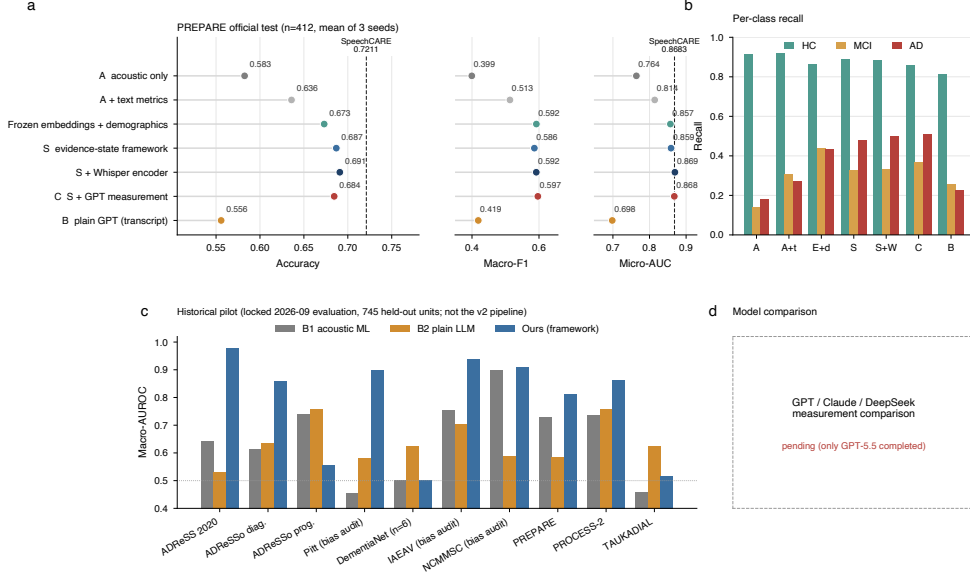


Fig. 3 Predictive results. (a) PREPARE official test set (n=412, mean of three seeds): accuracy, macro-F1 and micro-AUC for seven arms; dashed lines show published SpeechCARE means (accuracy 0.7211, micro-AUC 0.8683). All PREPARE test results are retrospective. (b) Per-class recall (A, acoustic; A+t, acoustic plus text; E+d, embeddings plus demographics; S, framework; S+W, framework plus Whisper; C, framework plus GPT measurement; B, plain GPT). (c) Historical pilot macro-AUROC across ten datasets (locked September 2026 evaluation, earlier pipeline). (d) Placeholder: the GPT, Claude and DeepSeek comparison is pending.

to 0.711 with nine; GPT measurement, demographics, text metrics, acoustic variables and states did not lower it (Fig. 4d). Many evidence groups are therefore redundant for prediction and are retained for traceability rather than accuracy.

Few-shot behaviour.

At 32 training participants the full framework reached test macro-AUC 0.659, compared with 0.633 for embeddings alone, 0.643 for states alone and 0.552 for acoustic features; at 128 participants the full framework and embeddings were similar (0.769 and 0.766), and at the full 1,622 the full framework reached 0.821 versus 0.807 for embeddings (Fig. 4e). Intervals at small sizes overlap, so we regard the few-shot advantage as suggestive.

Contamination.

A verbatim-continuation probe asked the LLM to continue the first half of transcripts. On public ADReSS CHAT transcripts (n=30), the mean 4-gram overlap with the true continuation was 0.007 versus 0.006 with another case's continuation, and no case reached 0.3. On PREPARE ASR transcripts, which were never published as text, overlap was 0.091 versus 0.030, with three of 30 cases at or above 0.3 (Fig. 4f). We found no evidence of memorization of the public corpus. Because the PREPARE transcripts

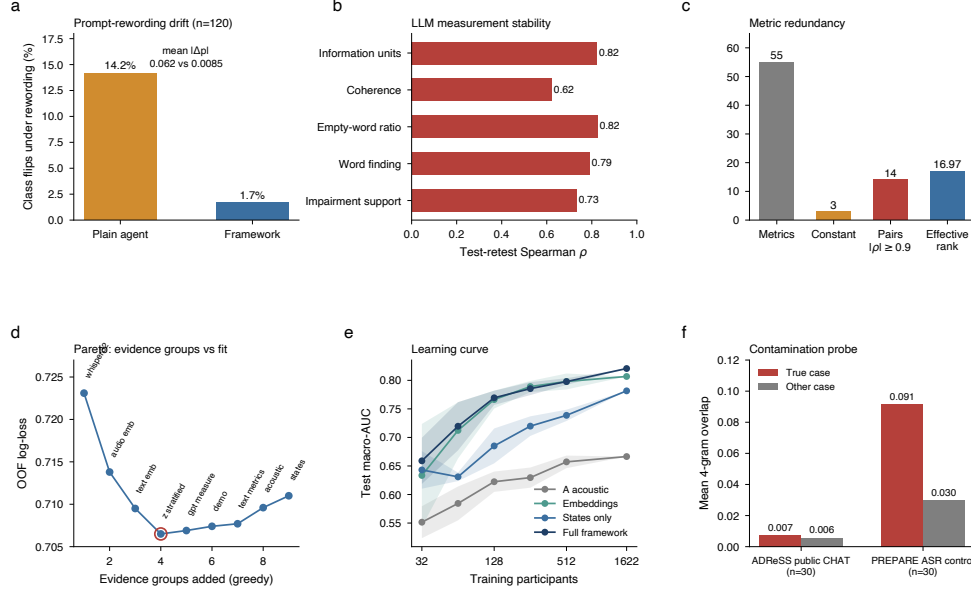


Fig. 4 Governance evidence. (a) Class flips and mean absolute probability change under prompt rewording (n=120). (b) Test–retest Spearman correlation of LLM measurements. (c) Metric redundancy: 55 metrics, three constant, 14 pairs with $|\rho| \geq 0.9$, effective rank 17.0. (d) Greedy evidence-group Pareto curve (out-of-fold log-loss). (e) Learning curves, test macro-AUC against training size (shaded: SD across subsamples). (f) Verbatim-continuation contamination probe, mean 4-gram overlap with the true versus another case’s continuation.

cannot have been memorized, overlap of this size arises without memorization, so the probe sets a baseline rather than excluding partial memorization.

5 Discussion

This study suggests that a cognitive-state layer can provide a useful intermediate interface between heterogeneous speech tasks and clinical screening outputs. Rather than forcing picture description, interview, structured task and spontaneous speech into a single undifferentiated feature vector, the framework first determines which states are observable in each task and how reliable each measurement is for the current case. On PREPARE, this organization improved discrimination over handcrafted acoustic features and a plain language model, while retaining a trace from the final report to the task, state, metric and local source segment. Most of the discrimination, however, came from pretrained encoders; the states did not add measurable AUC beyond them.

The contribution differs from SpeechCARE’s adaptive modality fusion. SpeechCARE learns how strongly a sample should rely on acoustic, linguistic and demographic representations and reaches higher accuracy on PREPARE (0.721 versus 0.691) while AUC is at parity (0.868 versus 0.869) [4]. Our framework instead constrains what those representations are allowed to mean clinically. Modality-level gating and cognitive-state evidence are complementary: the former is optimized for

predictive representation, whereas the latter specifies task observability, patient-level reliability, confounding and report permission. The present results do not show that interpretability compensates for lower predictive performance. Parity in AUC is not parity in accuracy, and the PREPARE test set was previously viewed; a comparison on untouched data under SpeechCARE’s repeated-run protocol is required before any equivalence claim.

The contribution also differs from MEDRF. MEDRF uses clinical profiles and structural MRI to predict a diagnostic hierarchy, then retrieves analogous cases and literature to refine ambiguous predictions [7]. Our hierarchy is a measurement pathway rather than a diagnostic ontology: local speech segments support metrics, metrics support cognitive states, and states support a screening probability. This design is intended for settings without imaging or complete neuropsychological profiles. However, MEDRF demonstrated actual prediction gains from RAG-LLM correction and obtained physician evaluations, whereas in our data LLM measurement gave no gain on ASR transcripts and the plain LLM was the weakest arm. The current evidence therefore supports an evidence-governed reporting and stability layer, not an autonomous diagnostic Agent.

The failure analyses are as important as the favourable datasets (Table 1). Progression prediction did not benefit from the cross-sectional state architecture, suggesting that longitudinal change requires within-person baselines and trajectory-specific supervision. TAUADIAL exposed weak transfer to Mandarin–English spontaneous speech, where ASR, language-specific lexical measurement and label definition differ from structured picture description. IAEAV, Pitt and NCMMSC showed that acquisition and quality variables can be highly predictive. Preventing these variables from appearing in the report is necessary, but it does not prevent correlated latent representations from exploiting them. Future evaluation should therefore use leave-site, leave-task and leave-language protocols, adversarial dataset-identity tests and prospective collection with balanced devices and interviewers.

Several limitations constrain clinical interpretation. This was a retrospective benchmark assembled from datasets with different labels and reference standards; it was not a prospective screening trial. The 745 held-out units should not be treated as one exchangeable clinical cohort. Some confidence intervals were wide, particularly for progression, IAEAV and DementiaNet. State definitions were literature-informed but have not yet undergone formal clinician content-validity assessment. The report evaluator was automated, and no blinded clinician comparison measured decision quality, review time or referral appropriateness. The Agent evidence validator rejected many candidates and applied no risk corrections. Finally, the framework predicts dataset labels for research screening; it cannot determine Alzheimer pathology, distinguish all alternative causes of cognitive impairment or recommend treatment.

Within these boundaries, the results establish a reproducible separation between predictive evidence, quality control and clinical communication. A clinically useful next version should combine SpeechCARE-level multilingual representation learning with the present state and provenance constraints, then evaluate whether clinicians can correct a state, observe a coherent probability change and make better referral

decisions. That experiment, rather than another increase in automatically generated report fluency, would test whether the evidence chain improves care.

Data availability

The study used datasets distributed under their respective source agreements. Several source recordings cannot be redistributed by the authors. Dataset adapters, inclusion rules, pseudonymous split identifiers and derived aggregate statistics will be released subject to those agreements. **[AUTHOR INPUT REQUIRED: list accession links, approval identifiers and the exact derived data that can be shared.]**

Code availability

The complete analysis code, configuration files, tests and report-generation workflow are maintained in the 8.27 project repository. **[AUTHOR INPUT REQUIRED: provide a public repository URL, immutable release tag, license and archival DOI before submission.]**

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